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LAB 7: Fuzzing for Input Validation Bugs

Screenshot 1: Execution of the Original Application

```
C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7>python train_fare.py
==== Running parse_pnr ====
Enter a 10 digit PNR (e.g. 4521367890): 4521367890
Zone and Number: (4, 521367890)

==== Running journey_distance ====
Enter station distances as CSV (e.g. 0,120,340): 0,120,340
Total Distance: 340

==== Running berth_type ====
Enter coach and seat (e.g. S3-45): S3-45
Coach and Berth: ('S3', 'MB')
```

Screenshot 2: Failing Fuzz Tests on the Original Application

```
C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7>python -m pytest test_train_fare.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0 -- C:\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7
plugins: anyio-4.14.0, hypothesis-6.170.15, cov-7.1.0
collected 3 items

test_train_fare.py::test_parse_pnr_no_crash FAILED                                                   [ 33%]
test_train_fare.py::test_journey_distance_no_crash FAILED                                            [ 66%]
test_train_fare.py::test_berth_type_no_crash FAILED                                                  [100%]
```

```
=================================================== FAILURES ===================================================
_____________________________________ test_parse_pnr_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 5, in test_parse_pnr_no_crash
  |     def test_parse_pnr_no_crash(s):
  |                    ^^^
  |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 7, in test_parse_pnr_no_crash
    |     parse_pnr(s)
    |     ~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 4, in parse_pnr
    |     return int(pnr[0]), int(pnr[1:])
    |                             ~~~^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 9, in test_parse_pnr_no_crash
    |     pytest.fail(f"parse_pnr crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: parse_pnr crashed with input '0': invalid literal for int() with base 10: ''
    | Failing test case: test_parse_pnr_no_crash(
    |     s='0',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 7, in test_parse_pnr_no_crash
    |     parse_pnr(s)
    |     ~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 4, in parse_pnr
    |     return int(pnr[0]), int(pnr[1:])
    |                         ~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 9, in test_parse_pnr_no_crash
    |     pytest.fail(f"parse_pnr crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: parse_pnr crashed with input '': string index out of range
    | Failing test case: test_parse_pnr_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 7, in test_parse_pnr_no_crash
    |     parse_pnr(s)
    |     ~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 4, in parse_pnr
    |     return int(pnr[0]), int(pnr[1:])
    |                         ~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 9, in test_parse_pnr_no_crash
    |     pytest.fail(f"parse_pnr crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: parse_pnr crashed with input None: 'NoneType' object is not subscriptable
    | Failing test case: test_parse_pnr_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
_____ test_journey_distance_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 11, in test_journey_distance_no_crash
  |     def test_journey_distance_no_crash(s):
  |                    ^^^
  |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 13, in test_journey_distance_no_crash
    |     journey_distance(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 7, in journey_distance
    |     distances = [int(d) for d in distances_csv.split(',')]
    |                  ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 15, in test_journey_distance_no_crash
    |     pytest.fail(f"journey_distance crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: journey_distance crashed with input '': invalid literal for int() with base 10: ''
    | Failing test case: test_journey_distance_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py:14
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 13, in test_journey_distance_no_crash
    |     journey_distance(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 7, in journey_distance
    |     distances = [int(d) for d in distances_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 15, in test_journey_distance_no_crash
    |     pytest.fail(f"journey_distance crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: journey_distance crashed with input None: 'NoneType' object has no attribute 'split'
    | Failing test case: test_journey_distance_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py:14
    +------------------------------------
```

```
_____ test_berth_type_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 17, in test_berth_type_no_crash
  |     def test_berth_type_no_crash(s):
  |                ^^^
  |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 19, in test_berth_type_no_crash
    |     berth_type(s)
    |     ~~~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 12, in berth_type
    |     return coach.upper(), BERTHS[int(number) % 8]
    |                               ~~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 21, in test_berth_type_no_crash
    |     pytest.fail(f"berth_type crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: berth_type crashed with input '-': invalid literal for int() with base 10: ''
    | Failing test case: test_berth_type_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 19, in test_berth_type_no_crash
    |     berth_type(s)
    |     ~~~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 11, in berth_type
    |     coach, number = seat.split('-')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 21, in test_berth_type_no_crash
    |     pytest.fail(f"berth_type crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: berth_type crashed with input '': not enough values to unpack (expected 2, got 1)
    | Failing test case: test_berth_type_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 19, in test_berth_type_no_crash
    |     berth_type(s)
    |     ~~~~~~~~~~~^^^
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\train_fare.py", line 11, in berth_type
    |     coach, number = seat.split('-')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7\test_train_fare.py", line 21, in test_berth_type_no_crash
    |     pytest.fail(f"berth_type crashed with input {s!r}: {e}")
    |     ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\prakr\AppData\Roaming\Python\Python314\site-packages\_pytest\outcomes.py", line 162, in __call__
    |     raise Failed(msg=reason, pytrace=pytrace)
    | Failed: berth_type crashed with input None: 'NoneType' object has no attribute 'split'
    | Failing test case: test_berth_type_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
========================================== short test summary info ==========================================
FAILED test_train_fare.py::test_parse_pnr_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_train_fare.py::test_journey_distance_no_crash - BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_train_fare.py::test_berth_type_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================================== 3 failed in 9.62s ==========================================
```

Screenshot 3: Execution of the Fixed Application.

```
C:\Users\prakr\Desktop\SEM 5\SE\experential learning\Experiential-learning-SE\Lab 7>python train_fare.py
==== Running parse_pnr ====
Enter a 10 digit PNR (e.g. 4521367890): 4521367890
Zone and Number: (4, 521367890)

==== Running journey_distance ====
Enter station distances as CSV (e.g. 0,120,340): 0,120,340
Total Distance: 340

==== Running berth_type ====
Enter coach and seat (e.g. S3-45): S3-45
Coach and Berth: ('S3', 'MB')
```

Screenshot 4: Passing Fuzz Tests After Fixing the Application

